

Fault-tolerant Cooperative Control of Multiple Quadrotor UAVs Handling a Common Load

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Abstract—This paper focuses on the modeling and a generalized control framework for the coordinated quadrotor UAVs-Load transportation system. At first, the relatively complete dynamic models for multiple quadrotor UAVs are established under undetectable actuation faults and uncertain wind interference, considering the external effects of the load on the translational and rotational motions of each quadrotor UAV. And in order to transport the load smoothly and safely, the consensus control problem how to make the UAVs maintain the designed geometric configuration is formulated correspondingly. Finally, by introducing the virtual parameter estimation error into the Lyapunov function based on neighboring UAVs state information, the adaptive fault-tolerant controllers are developed to ensure that the defined synchronization errors for the UAVs position and attitude are uniform ultimate bound(UUB). The simulation results also prove the effectiveness of the proposed control approach.

Index Terms—Load transportation, quadrotor UAVs, actuation faults, fault-tolerant control, consensus control

I. Introduction

Quadrotor unmanned aerial vehicles (UAVs) have found extensive application in both military and civilian domains, including but not limited to, enemy detection, environmental monitoring, cargo delivery, power inspection, and agricultural protection [1] [2]. Accomplishing these tasks necessitates that the quadrotor UAVs be capable of carrying a certain weight of equipment or goods. In cases where the weight of the cargo exceeds the payload capacity of single aircraft, multiple quadrotor UAVs must be employed for cooperative transportation. The successful execution of multi-UAVs cooperative carrying tasks not only requires that each UAV maintain stable flight ability despite the influence of load weight, but also poses a significant challenge to the design of the cooperative control strategy [3].

System modeling and controller design for quadrotor UAV carrying a load in flight have been extensively studied recently, as they are essential components of multi-UAVs collaborative load transportation research [4]. To cater to different missions and object types, various connection

means between UAV and load, such as magnetic attractors, robotic manipulators, and suspension ropes or cables, have been explored. As a result, simple and convenient cables are frequently used to hang the cargo on quadrotor UAV. Using the method of cable connection, a quadrotor dynamic model and a trajectory tracking controller that are suitable for cable loads are established by Lagrange equations [5]. Moreover, the quality and elasticity of the cable connecting the UAV and the load cannot be ignored during the modeling process. In literature [6], a rigid cable is used to suspend the load on a quadrotor, and the dynamics model of the UAV-load position subsystem and the UAV attitude subsystem are established. Additionally, a trajectory tracking controller is designed for the position of the point-mass load.

As single UAV-load flight system become more prevalent, recent research has begun to explore the integration of multi-agent formation theory into the control of collaborative loading transportation with two or more quadrotor UAVs [7] [8]. From a load shape and number of aircraft perspective, a typical quadrotor UAVs-load system can be categorized into six cases where two or more aircraft carry a mass-point load, rod-shaped goods, or generic geometric cargo. Cooperative control of multiple UAVs is necessary to achieve position and attitude control of the load. Specifically, controlling the pose of a rod-shaped load requires at least two UAVs, while controlling the pose of a regular geometric cargo requires at least three UAVs [9]. Literature [10] incorporates coupling dynamics between rigid body payload, cables and quadrotors into the design of the control system, and presents a control system for transporting geometric loads of an arbitrary number of quadrotors. Obviously, the above literatures focus on the number of UAVs and load type. Most studies have not quantified the impact of suspended load on each UAV, and adopt the design of fixed expectation formation. Only a few studies have considered the load weight distribution in the UAVs flight, which highlights the potential to combine multi-agent formation control with load transportation missions.

In this work, we consider the consensus control problem of leader-follower UAVs with distributed load effect,

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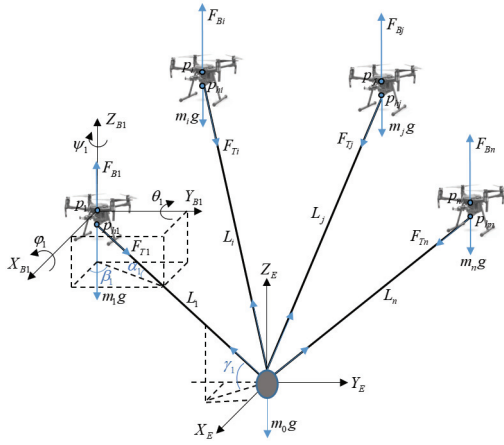


Fig. 1. Schematic of quadrotor UAVs-Load transportation system

unknown actuation faults, and uncertain disturbances. The main contributions of this work can be summarized following.

1) In view of the undetectable actuation faults and uncertain wind interference, a dynamic model of synchronous error in consensus state for transporting mass-point load of multiple quadrotor UAVs is established.

2) The consensus control problem of how to make the UAVs maintain the designed geometric configuration is formulated correspondingly, so as to transport the load smoothly and safely.

3) Based on the status information of neighboring UAVs, an adaptive fault-tolerant control strategy is proposed to solve the trajectory tracking problem of the suspended load UVAs under actuator faults. The effectiveness of the proposed method is verified by Lyapunov stability analysis and Matlab simulation.

II. System Modeling and Problem Formulation

A. System dynamical modeling with load effect and actuation faults

As shown in Fig.1, we first define the inertial frame $O_E = (X_E, Y_E, Z_E)$, and the body frame $O_{Bi} = (X_{Bi}, Y_{Bi}, Z_{Bi})$ for the i -th quadrotor UAV.

Assumption 1. The quadrotor UAVs are rigid and symmetrical, then the center of gravity of quadrotor UAV itself is coincides with its geometric center.

Assumption 2. The load is suspended on the i -th quadrotor UAV through the i -th cable L_i , the load is considered as a mass point, and the cables are assumed as massless and inelastic with the same length L .

Assumption 3. The load suspension point on the i -th quadrotor UAV is located directly below its geometric center, i.e., the load suspension point can be expressed by $p_{hi} = [0, 0, -h]^T$ in the frame O_{Bi} , where $h > 0$.

The position of the load in the frame O_E is denoted as $p_0 = [x_0, y_0, z_0]^T$, then we have $p_0 = p_i + R_{Bi}^E p_{B0i}$, where p_{B0i}

is the load position in the frame O_{Bi} and will be given later.

Combined with Assumption 2, the load p_0 is located at the center of a sphere determined by n quadrotor UAVs' load suspension points $p_{hi} = [x_{hi}, y_{hi}, z_{hi}]^T$ in the frame O_E and the cable L_i is one of the sphere's radii, so p_0 can be calculated by $p_0 = \bar{p}_0 + \bar{r}\bar{h}$, where $\bar{p}_0$ is the circumcenter of the triangle determined by three UAVs' suspension points $\{p_i, p_j, p_k | 1 \leq i < j < k \leq n\}$, the detail calculation process of $\bar{p}_0$ is omitted here, $\bar{r} = \frac{(p_{hj}-p_{hk}) \times (p_{hk}-p_{hi})}{\|(p_{hj}-p_{hi}) \times (p_{hk}-p_{hi})\|}$ denotes the unit direction vector from $\bar{p}_0$ to the load p_0 , $\bar{h} = \sqrt{L^2 - \|\bar{p}_0 - p_{hi}\|^2}$ is the distance between $\bar{p}_0$ and the load p_0 .

To facilitate the description of the impact of load on UAVs' flight states, we introduce two azimuth angles α_i and β_i as shown in Fig.1, where α_i is the angle between the cable L_i and the plane $X_{Bi}O_{Bi}Y_{Bi}$, and β_i is defined as the angle of the projection line of the cable L_i on the plane $X_{Bi}O_{Bi}Y_{Bi}$. So we know $0 < \alpha_i \leq \frac{\pi}{2}$ and $-\frac{\pi}{2} \leq \beta_i \leq \frac{3\pi}{2}$.

Hence, the load position p_{B0i} can be expressed with

$$p_{B0i} = p_{hi} + r_{Bli} \quad (1)$$

$$r_{Bli} = [L_i C \alpha_i C \beta_i, L_i C \alpha_i S \beta_i, -L_i S \alpha_i]^T \quad (2)$$

where r_{Bli} is the direction vector from the suspension point p_{hi} to the load position p_{B0i} in the frame O_{Bi} , which can be calculated with $r_{Bli} = (R_{Bi}^E)^{-1}(p_0 - p_{hi}) = [x_{Bli}, y_{Bli}, z_{Bli}]^T$.

Next, in the frame O_{Bi} , the absolute acceleration a_{B0i} can be expressed based on the i -th UAVs' states as follows [17]

$$a_{B0i} = (R_{Bi}^E)^{-1} \ddot{p}_i + \dot{w}_i \times p_{B0i} + w_i \times (w_i \times p_{B0i}) + \ddot{p}_{B0i} + 2w_i \times \dot{p}_{B0i} \quad (3)$$

In this work, we calculate the pulling force F_{ti} and the pulling moment M_{ri} as below

$$F_{ti} = -\Lambda_i (G_0 + m_0 R_{Bi}^E a_{B0i}) \quad (4)$$

$$M_{ri} = p_{Bhi} \times ((R_{Bi}^E)^{-1} \cdot F_{ti}) \quad (5)$$

where the direction of F_{ti} is from the i -th UAV to the load, $\Lambda_i = \frac{\gamma_i}{\sum_{i=1}^n \gamma_i}$ denote the distribution coefficient of load gravity, $\sum_{i=1}^n \Lambda_i = 1$, γ_i is defined as the angle between the cable L_i and the plane $X_E O_E Y_E$ of the frame O_E , and $\gamma_i = \arcsin(\frac{z_{hi} - z_0}{L_i})$.

Therefore, the dynamic model of the UAVs' translation and rotation subsystems can be rewritten as

$$\bar{A}_i \dot{v}_i = u_{ti} + G_i + \bar{F}_{ti} + d_{ti}(\cdot) \quad (6)$$

$$\bar{B}_i \dot{w}_i = u_{ri} + \bar{\Omega}_i + \bar{M}_{ri} + d_{ri}(\cdot) \quad (7)$$

where $\bar{A}_i = \text{diag}\{m_i + \Lambda_i m_0, m_i + \Lambda_i m_0, m_i + \Lambda_i m_0\}$, $\bar{B}_i = \begin{bmatrix} J_{xi} + \Delta_{i1} & \Delta_{i2} \\ J_{yi} + \Delta_{i1} & \Delta_{i3} \\ J_{zi} & \Delta_{i3} \end{bmatrix}$, $\Delta_{i1} = -\Lambda_i m_0 h z_{B0i} > 0$, $\Delta_{i2} = \Lambda_i m_0 h x_{B0i}$, $\Delta_{i3} = -\Lambda_i m_0 h y_{B0i}$, it's easy to know the moment of inertia matrix $\bar{B}_i$ are invertible. $\bar{\Omega}_i = \bar{B}_i^{-1} \bar{w}_i$, $\bar{F}_{ti}$ is the

remaining item of F_{ri} that does not contain $\dot{v}_i, \bar{M}_{ri}$ is the remaining item of M_{ri} that does not contain $\dot{w}_i$ and w_i . $d_{li}(\cdot)$ and $d_{ri}(\cdot)$ denote the external interference produced by the wind resistance and the coupling of airflow among UAVs.

Assumption 4. The external wind interference items $d_{li}(\cdot)$ and $d_{ri}(\cdot)$ are unknown, time-varying and bounded, i.e., there exists unknown constants $\bar{d}_{li}$ and $\bar{d}_{ri}$, such that $\{|d_{xi}(\cdot)|, |d_{yi}(\cdot)|, |d_{zi}(\cdot)|\} \leq \bar{d}_{li} < \infty$ and $\{|d_{\phi i}(\cdot)|, |d_{\theta i}(\cdot)|, |d_{\psi i}(\cdot)|\} \leq \bar{d}_{ri} < \infty$.

Once actuation fault occurs, the actual UAVs' control outputs will no longer be consistent with the control inputs u_{ti} and u_{ri} , the corresponding relationship as follows

$$u_{ti} \Rightarrow \rho_{ti}(\cdot)u_{ti} + \delta_{ti}(\cdot), t \geq t_{Ti} \quad (8)$$

$$u_{ri} \Rightarrow \rho_{ri}(\cdot)u_{ri} + \delta_{ri}(\cdot), t \geq t_{Ri} \quad (9)$$

where t_{Ti} and t_{Ri} are the times instant at which the actuation fault happens, $\rho_{ti}(\cdot) = \text{diag}\{\rho_{xi}(\cdot), \rho_{yi}(\cdot), \rho_{zi}(\cdot)\}$ and $\rho_{ri}(\cdot) = \text{diag}\{\rho_{\phi i}(\cdot), \rho_{\theta i}(\cdot), \rho_{\psi i}(\cdot)\}$ denote the actuation effectiveness of the i -th quadrotor UAV, $\delta_{ti}(\cdot) = [\delta_{xi}(\cdot), \delta_{yi}(\cdot), \delta_{zi}(\cdot)]^T$ and $\delta_{ri}(\cdot) = [\delta_{\phi i}(\cdot), \delta_{\theta i}(\cdot), \delta_{\psi i}(\cdot)]^T$ represent the uncontrollable output part produced by the i -th actuator action.

Hence, after considering the actuation faults, the translational motion subsystem (17) and the rotational motion subsystem (18) are further rewritten as:

$$\bar{A}_i \dot{v}_i = \rho_{ti}(\cdot)u_{ti} + G_i + \bar{F}_{ti} + d_{li}(\cdot) + \delta_{ti}(\cdot) \quad (10)$$

$$\bar{B}_i \dot{w}_i = \rho_{ri}(\cdot)u_{ri} + \bar{\Omega}_i + \bar{M}_{ri} + d_{ri}(\cdot) + \delta_{ri}(\cdot) \quad (11)$$

Assumption 5. The actuation effectiveness functions $\rho_{ti}(\cdot)$, $\rho_{ri}(\cdot)$ and the actuator additional output function $\delta_{ti}(\cdot)$, $\delta_{ri}(\cdot)$ are unknown, time-varying and bounded, there exist unknown constants $\{\rho_{im}, \rho_{rm}, \bar{\delta}_{ti}, \bar{\delta}_{ri}\}$, so that $0 < \rho_{im} \leq \{\rho_{xi}(\cdot), \rho_{yi}(\cdot), \rho_{zi}(\cdot)\} \leq 1$, $0 < \rho_{rm} \leq \{\rho_{\phi i}(\cdot), \rho_{\theta i}(\cdot), \rho_{\psi i}(\cdot)\} \leq 1$, $\{|\delta_{xi}(\cdot)|, |\delta_{yi}(\cdot)|, |\delta_{zi}(\cdot)|\} < \bar{\delta}_{ti} < \infty$, $\{|\delta_{\phi i}(\cdot)|, |\delta_{\theta i}(\cdot)|, |\delta_{\psi i}(\cdot)|\} < \bar{\delta}_{ri} < \infty$.

B. Multi-UAVs formation model considering load weight distribution

Assumption 6. The angle Υ between the projection lines of the two cables on the i th and $i+1$ th UAV on the $X_E O_E Y_E$ plane is the same, satisfying $\Upsilon = 2\pi/n$.

Assumption 7. The position of the projection line of the leader UVA's cable on the $X_E O_E Y_E$ plane parallel to the X_E axis is selected as the initial time position.

The load distribution angle γ_i is proportional to its load distribution coefficient Λ_i , so we can get $\gamma_i = \gamma_1(\Lambda_i/\Lambda_1)$ and $\sum_{i=1}^n \Lambda_i = 1$ is satisfied. Given the expected load motion trajectory p_0^* and expected load distribution angle of leader UAV γ_1 , combining with the above assumptions, the expected formation model of multiple UAVs can be displayed and expressed.

Lemma 1. If several UAVs are working together to transport a load, the expected formation model for the

weight distribution can be achieved through adjusting the position of each UAV relative to the payload as shown in $p_i^* = p_0^* - R_{B1}^E p_{Bhi} + [Lcos(\gamma_i)cos(-(i-1)\Upsilon), Lcos(\gamma_i)sin(-(i-1)\Upsilon), Lsin(\gamma_i)]^T$, where $\Upsilon = 2\pi/n$ and $\gamma_i = \gamma_1(\Lambda_i/\Lambda_1)$.

Proof: Given the expected load distribution coefficient of UAVs, taking the cooperative load system of three UAVs as an example, when considering the above assumptions, there is $\Upsilon = 2\pi/3$, so $\gamma_2 = \gamma_1(\Lambda_2/\Lambda_1)$ and $\gamma_3 = \gamma_1(\Lambda_3/\Lambda_1)$ are obtained. The top view of it under the O_E system is shown in the figure 2.

For No. 1 leader UAV, its desired flight trajectory is $p_1^* = p_0^* - R_{B1}^E p_{Bh1} - r_{l1} = p_0^* - R_{B1}^E p_{Bh1} - [-a_1, -b_1, -h_1]^T$ by calculating $h_1 = Lsin(\gamma_1)$, $r_1 = Lcos(\gamma_1)$, $a_1 = r_1$ and $b_1 = 0$. For No. 2 follower UAV, by calculating $h_2 = Lsin(\gamma_2)$, $r_2 = Lcos(\gamma_2)$, $a_2 = r_2cos(-\Upsilon)$ and $b_2 = r_2sin(-\Upsilon)$ its desired flight trajectory is obtained as $p_2^* = p_0^* - R_{B2}^E p_{Bh2} - [-a_2, -b_2, -h_2]^T$ similarly. For No. 3 follower UAV, by calculating $h_3 = Lsin(\gamma_3)$, $r_3 = Lcos(\gamma_3)$, $a_3 = r_3cos(-2\Upsilon)$ and $b_3 = r_3sin(-2\Upsilon)$, the expected flight trajectory is $p_3^* = p_0^* - R_{B3}^E p_{Bh3} - [-a_3, -b_3, -h_3]^T$ in the same way.

Finally, the multi-aircraft expected formation model of n UAVs payload system can be obtained by analogy as follows

$$\begin{aligned} p_i^* &= p_0^* - R_{Bi}^E p_{Bhi} - r_{li} = p_0^* - R_{Bi}^E p_{Bhi} - [-a_i, -b_i, -h_i]^T \\ &= p_0^* - R_{Bi}^E p_{Bhi} + [Lcos(\gamma_i)cos(-(i-1)\Upsilon), \\ &\quad Lcos(\gamma_i)sin(-(i-1)\Upsilon), Lsin(\gamma_i)]^T \end{aligned} \quad (12)$$

where $h_i = Lsin(\gamma_i)$, $a_i = r_i cos(-(i-1)\Upsilon)$ and $b_i = r_i sin(-(i-1)\Upsilon)$.

C. Problem formulation and the control objective

Based on the theory of multi-agent consistency, the synchronization errors between the i -th UAV and its neighbor UAVs in the position loop and attitude loop are defined as e_{ti} and e_{ri} respectively as follows

$$e_{ti} = \sum_{j \in N_i} a_{ij} [(p_{ti} - p_{tj}) - p_{tij}^*] \quad (13)$$

$$e_{ri} = \sum_{j \in N_i} a_{ij} [(q_{ri} - q_{rj}) - q_{rij}^*] \quad (14)$$

where $p_{tij}^* = p_{ti}^* - p_{tj}^*$ denotes the designed relative position, $q_{rij}^* = 0$ indicates that the attitude angles of the i -th UAV and the j -th UAV are expected to be consistent.

To facilitate the controllers' design, we introduce the intermediate error variable s_{ti} , ξ_{ti} , s_{ri} , η_i as below

$$s_{ti} = \dot{e}_{ti} + \beta_t e_{ti}, \quad \xi_{ti} = (\dot{p}_i - \dot{p}_i^*) + \beta_t (p_i - p_i^*) \quad (15)$$

$$s_{ri} = \dot{e}_{ri} + \beta_r e_{ri}, \quad \eta_i = \dot{q}_i + \beta_r q_i \quad (16)$$

So when the quadrotor UAVs is consider the presence of actuation faults, combining (10) and (11), the synchronization errors' dynamics can be written as follows

$$\dot{\xi} = \bar{P} + \beta_t \bar{P} = A_t^{-1}(\rho_t U_t + G_t + L_t^1) - \dot{V}_t^* + \beta_t V_t - \beta_t V_d^* \quad (17)$$

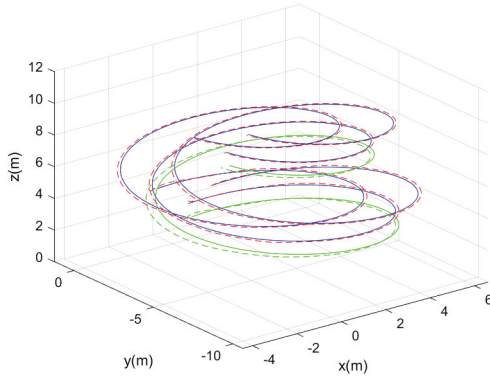


Fig. 2. Simulation performance of UAVs-Load system under actuation faults

$$\dot{\eta} = \bar{Q} + \beta_r \bar{Q} = B_r^{-1}(\rho_r U_r + \Omega_r + L_r^1) - \dot{W}_r^* + \beta_r W_r - \beta_r W_r^* \quad (18)$$

where $L_t^1 = [L_{t1}^1, \dots, L_{tn}^1]^T$ and $L_r^1 = [L_{r1}^1, \dots, L_{rn}^1]^T$ are the lumped uncertainties of the UAVs' translation and rotation respectively, with $L_{t1}^1 = \bar{F}_{ti} + d_{ti}(\cdot) + \delta_{ti}(\cdot) = \{L_{xi}^1, L_{yi}^1, L_{zi}^1\}$ and $L_{ri}^1 = \bar{\Omega}_i + d_{ri}(\cdot) + \delta_{ri}(\cdot) = \{L_{\phi i}^1, L_{\theta i}^1, L_{\psi i}^1\}$.

The objective in this paper is to design adaptive fault-tolerant controllers such that the undetectable actuation faults and uncertain wind perturbations could be compensated, the synchronization errors S_t and S_r converge to the residual sets near the origin. Meanwhile, the system is stable in the sense of uniform ultimate bound(UUB) and all internal signals are bounded.

III. Controller Design And Stability Analysis

Consider the system synchronization errors' dynamic model (17) and (18), if the following control scheme

$$u_{ti} = -k_{ti} s_{ti} + u_{ai}, \quad u_{ai} = -\frac{\hat{c}_{ti} \zeta_{ti}^2 s_{ti}}{\zeta_{ti} |s_{ti}| + \tau} \quad (19)$$

$$u_{ri} = -k_{ri} s_{ri} + u_{bi}, \quad u_{bi} = -\frac{\hat{c}_{ri} \zeta_{ri}^2 s_{ri}}{\zeta_{ri} |s_{ri}| + \tau} \quad (20)$$

where $k_{ti} > 0$ and $k_{ri} > 0$ are the designed constants, ζ_{ti} and ζ_{ri} represent the non-negative known scalar functions to be given later, $\hat{c}_{ti}$ and $\hat{c}_{ri}$ are the estimations of the virtual parameters c_{ti} and c_{ri} respectively, c_{ti} and c_{ri} are defined later, the adaptive law as

$$\dot{\hat{c}}_{ti} = -\sigma_{ti} \hat{c}_{ti} + \frac{\sigma_{ti} \zeta_{ti}^2 s_{ti}^2}{\zeta_{ti} |s_{ti}| + \tau}, \quad \dot{\hat{c}}_{ri} = -\sigma_{ri} \hat{c}_{ri} + \frac{\sigma_{ri} \zeta_{ri}^2 s_{ri}^2}{\zeta_{ri} |s_{ri}| + \tau} \quad (21)$$

where $\{\sigma_{ti}, \sigma_{ri}\}$ are positive design parameters, the following control objectives under actuation faults condition can be achieved

- 1) the state synchronization errors S_t and S_r converges to a small set near origin;
- 2) all system signals are bounded.

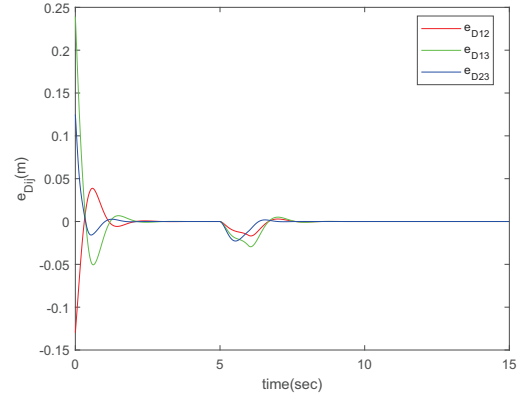


Fig. 3. Formation errors between UAV_i and UAV_j under actuation faults

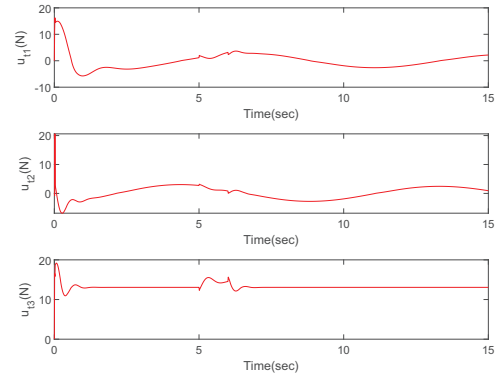


Fig. 4. Control inputs of UAVs' translational motions under actuation faults

Proof: First, we consider the translation subsystem corresponding to the synchronization error dynamics 17. Since the weighted graph G is undirected and connected, there exists an orthogonal matrix U_n such that $\mathcal{L} = U_n^T \text{diag}\{0, \lambda_2, \dots, \lambda_n\} U_n = U_n^T T U_n$ with $T = \text{diag}\{0, \lambda_2, \dots, \lambda_n\}$. Meanwhile, define $\bar{T} = \text{diag}\{c, \lambda_2, \dots, \lambda_n\}$ and $R = U_n^T \bar{T}^{-1} U_n$, c is a small positive constant. So we have $T \bar{T}^{-1} T = T$, and R is positive definite symmetric matrix. Consider the first part of Lyapunov candidate function V_1 as

$$V_1 = \frac{1}{2} S_t^T R S_t \quad (22)$$

Then combined with (17), the time derivative of V_1 is expressed by

$$\begin{aligned} \dot{V}_1 &= S_t^T R \dot{S}_t = \xi_t^T \mathcal{L}^T (U_n^T \bar{T}^{-1} U_n) \mathcal{L} \xi_t = S_t^T \dot{\xi}_t \\ &= -S_t^T A_t^{-1} \rho_t K_t S_t + S_t^T (A_t^{-1} (\rho_t U_a + G_t + L_t^1) - \dot{V}_t^* \\ &\quad + \beta_t V_t - \beta_t V_t^*) \end{aligned} \quad (23)$$

where $K_t = \text{diag}\{k_{t1}, \dots, k_{tn}\}$ is a positive definite diagonal matrix with respect to the positive parameter k_{ti} to be

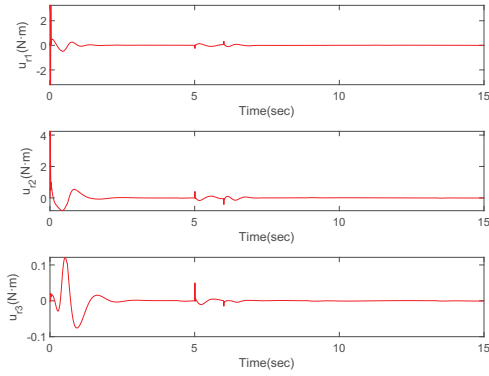


Fig. 5. Control inputs of UAVs' rotational motions under actuation faults

designed, $\rho_t = \text{diag}\{\rho_{t1}, \dots, \rho_{tn}\}$ is a positive definite diagonal matrix with respect to $0 < \rho_{ti} < 1$, $U_a = [u_{a1}, \dots, u_{an}]^T$.

Combining Assumption 4 and Assumption 5, there exists a positive constant $\bar{L}_{ii}^1$, such that the lumped uncertain item satisfies $|L_{ii}^1| \leq \bar{L}_{ii}^1$. Hence, we have $|(m_i + \Lambda_i m_0)^{-1}(G_{ii} + L_{ii}^1) - \dot{v}_{ii}^* + \beta_t v_{ti} - \beta_t v_{ii}^*| \leq c_{ti} \zeta_{ii}$, where c_{ti} is a non-negative unknown constant defined as $c_{ti} = \max\{1, \beta_t, \bar{L}_{ii}^1\} < \infty$, and ζ_{ii} is a non-negative known scalar function defined as $\zeta_{ii}(\cdot) = |(m_i + \Lambda_i m_0)^{-1}G_{ii} - \dot{v}_{ii}^* + (m_i + \Lambda_i m_0)^{-1}| + |v_{ti} - v_{ii}^*|$. Then $\dot{V}_1$ can be rewritten by

$$\begin{aligned} \dot{V}_1 &\leq S_t^T A_t^{-1} \rho_t K_t S_t + \sum_{i=1}^n \left(\frac{\rho_{tm}}{-(m_i + \Lambda_i m_0)} \frac{\hat{c}_{ti} \zeta_{ii}^2 s_{ii}^2}{\zeta_{ii} |s_{ii}| + \tau} + c_{ti} \zeta_{ii} |s_{ii}| \right) \\ &= -S_t^T A_t^{-1} \rho_t K_t S_t + \sum_{i=1}^n \left(\frac{\rho_{tm}}{-(m_i + \Lambda_i m_0)} \frac{\hat{c}_{ti} \zeta_{ii}^2 s_{ii}^2}{\zeta_{ii} |s_{ii}| + \tau} \right) \\ &\quad + \sum_{i=1}^n \frac{c_{ti} \zeta_{ii}^2 s_{ii}^2 + c_{ti} \zeta_{ii} \tau |s_{ii}|}{\zeta_{ii} |s_{ii}| + \tau} \\ &\leq -S_t^T A_t^{-1} \rho_t K_t S_t + \sum_{i=1}^n \left((c_{ti} - \frac{\rho_{tm}}{m_i + \Lambda_i m_0} \hat{c}_{ti}) \frac{\zeta_{ii}^2 s_{ii}^2}{\zeta_{ii} |s_{ii}| + \tau} + c_{ti} \tau \right) \end{aligned} \quad (24)$$

where $\frac{\zeta_{ii} |s_{ii}|}{\zeta_{ii} |s_{ii}| + \tau} \leq 1$ is used and $\rho_{tm} = \min\{\rho_{t1}, \dots, \rho_{tn}\}$.

Next, here we define the virtual parameter estimation error $\tilde{c}_{ti}$ as $\tilde{c}_{ti} = c_{ti} - \frac{\rho_{tm} \hat{c}_{ti}}{m_i + \Lambda_i m_0}$.

So we introduce the second part of Lyapunov function candidate V_2 about the adaptive parameter c_{ti} as

$$V_2 = \frac{1}{2} \frac{m_i + \Lambda_i m_0}{\rho_{tm}} \tilde{C}_t^T \Gamma_t^{-1} \tilde{C}_t \quad (25)$$

where $\Gamma_t = \text{diag}\{\sigma_{t1}, \dots, \sigma_{tn}\}$ is a positive definite diagonal matrix with respect to the positive parameter σ_{ti} to be designed, and $\tilde{C}_t = [\tilde{c}_{t1}, \dots, \tilde{c}_{tn}]^T$.

And using the adaptive law $\dot{\hat{c}}_{ti}$ given by (21), the time derivative of V_2 is expressed by

$$\begin{aligned} \dot{V}_2 &= \tilde{C}_t^T \Gamma_t^{-1} (-\dot{\tilde{C}}_t) = \sum_{i=1}^n \left((c_{ti} - \frac{\rho_{tm}}{m_i + \Lambda_i m_0} \hat{c}_{ti}) (-\frac{\dot{\hat{c}}_{ti}}{\sigma_{ti}}) \right) \\ &= \sum_{i=1}^n \left((c_{ti} - \frac{\rho_{tm}}{m_i + \Lambda_i m_0} \hat{c}_{ti}) (\hat{c}_{ti} - \frac{\zeta_{ii}^2 s_{ii}^2}{\zeta_{ii} |s_{ii}| + \tau}) \right) \end{aligned} \quad (26)$$

By choosing the Lyapunov function candidate $V_a = V_1 + V_2$, combining (24) and (26), we have

$$\dot{V}_a = \dot{V}_1 + \dot{V}_2 \leq -S_t^T A_t^{-1} \rho_t K_t S_t + \sum_{i=1}^n \tilde{c}_{ti} \hat{c}_{ti} + \sum_{i=1}^n c_{ti} \tau \quad (27)$$

Resort to the inequality $\tilde{c}_{ti} \hat{c}_{ti} \leq \frac{1}{2} (m_i + \Lambda_i m_0) \rho_{tm}^{-1} (c_{ti}^2 - \tilde{c}_{ti}^2)$, then (27) can be rewritten as

$$\begin{aligned} \dot{V}_a &\leq -S_t^T A_t^{-1} \rho_t K_t S_t - \frac{1}{2} (m_i + \Lambda_i m_0) \rho_{tm}^{-1} \sum_{i=1}^n \tilde{c}_{ti}^2 \\ &\quad + \frac{1}{2} (m_i + \Lambda_i m_0) \rho_{tm}^{-1} \sum_{i=1}^n c_{ti}^2 + \sum_{i=1}^n c_{ti} \tau \end{aligned} \quad (28)$$

There exists $-S_t^T A_t^{-1} \rho_t K_t S_t \leq -(m_i + \Lambda_i m_0)^{-1} \rho_{tm} k_{tm} S_t^T S_t$ with $k_{tm} = \min\{k_{t1}, \dots, k_{tn}\}$, so we get

$$\dot{V}_a \leq -l_1 V_a + l_2 \quad (29)$$

where $l_1 = \min\{2(m_i + \Lambda_i m_0)^{-1} \rho_{tm} k_{tm} \lambda_{\max}^{-1}(R), \lambda_{\max}^{-1}(\Gamma_t)\}$, $l_2 = \frac{1}{2} (m_i + \Lambda_i m_0) \rho_{tm}^{-1} \sum_{i=1}^n c_{ti}^2 + \sum_{i=1}^n c_{ti} \tau$.

Thus, we can get that $V_a \in L_\infty$, then $S_t \in L_\infty$ and $\tilde{C}_t \in L_\infty$ are ensured, hence $U_t \in L_\infty$.

The stability proof process of the rotation subsystem is similar to that of the translation subsystem, so it's omitted.

IV. Simulation

The actuation faults $\rho_t = [0.8, 0.7, 0.9]^T$, $\rho_r = [0.8, 0.7, 0.9]^T$, and $\delta_t = [2\cos(0.1t)\sin(0.5t), 2\sin(0.2t)\cos(0.2t), 0.03]^T$, $\delta_r = [2\cos(0.1t)\sin(0.5t), 2\sin(0.1t)\cos(0.3t), 0.04]^T$, and the corresponding controllers' parameters are assigned with $\beta_t = [3, 3, 2]^T$, $\beta_r = [3, 3, 2]^T$, $k_t = [3, 3, 3]^T$, $k_r = [3, 3, 3]^T$, $\sigma_t = [1, 1, 1]^T$, $\sigma_r = [1, 1, 1]^T$. Calculate $\Lambda_1 = \rho_{t1}/(\rho_{t1} + \rho_{t2} + \rho_{t3}) = 0.33$, $\Lambda_2 = 0.29$ and $\Lambda_3 = 0.38$. Given the expected motion trajectory of the load as $p_0^* = [0.9 + 5\sin(0.7t), 5\cos(0.7t) - 5, 1.4 + 0.5t]^T$, set the expected load distribution angle of the leading UAV as $\gamma_1 = 60^\circ$. $p_1^* = [1.9 + 5\sin(0.7t), 5\cos(0.7t) - 5, 3.2 + 0.5t]^T$, $p_2^* = [0.3 + 5\sin(0.7t), 5\cos(0.7t) - 6.2, 3.1 + 0.5t]^T$ and $p_3^* = [0.5 + 5\sin(0.7t), 5\cos(0.7t) - 4.3, 3.4 + 0.5t]^T$. The desired formation trajectory for transporting load by coordinating multiple quadrotor UAVs and the actual formation trajectory can be seen in Fig. 3. In addition, Fig. 4 represents the detail of the varying formation as expected. The translation force control inputs u_{ti} and the rotation moment control inputs u_{ri} of the three quadrotor UAVs are displayed in Fig. 5 and Fig. 6 respectively. And Fig. 7 and Fig. 8 show the translation and the rotation tracking performances under the u_{ti} and u_{ri} , we could see that the three quadrotor UAVs are also controlled well with realizable amplitude ranges and acceptable tracking errors after the designed actuators faults take into account.

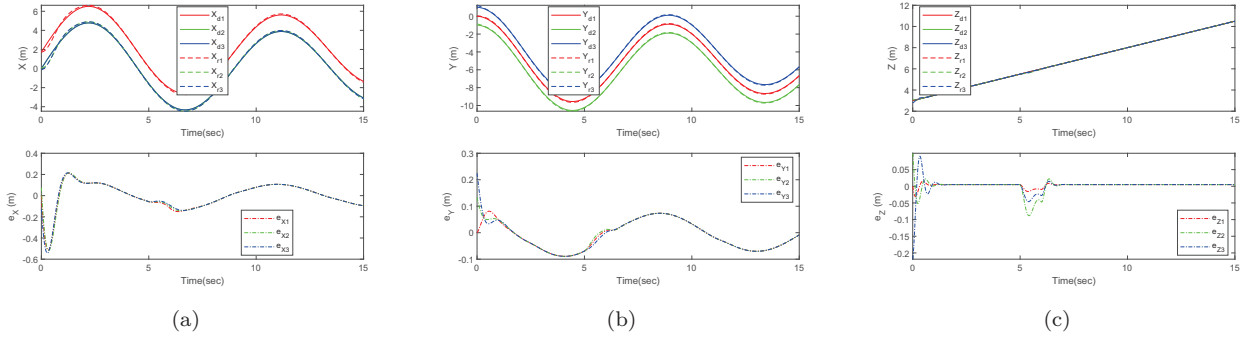


Fig. 6. The UAVs' translation tracking errors under actuation faults. (a)the X_i 's tracking errors. (b)the Y_i 's tracking errors. (c)the Z_i 's tracking errors.

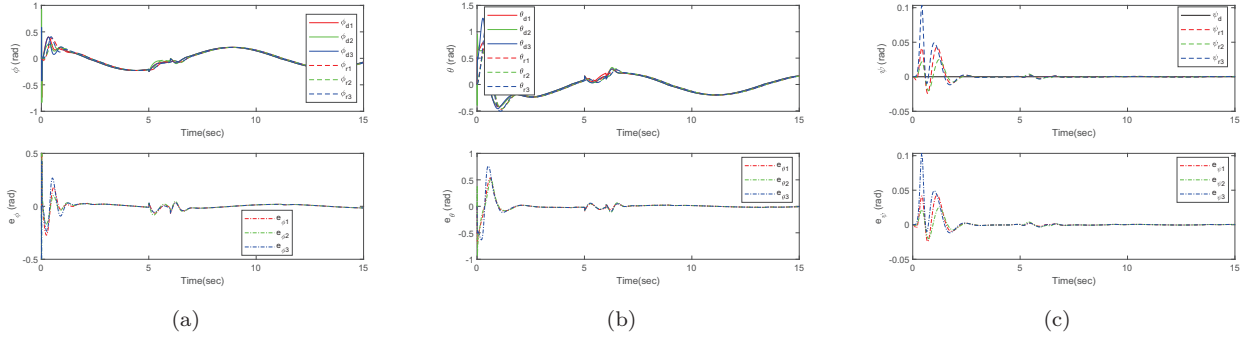


Fig. 7. The UAVs' rotation tracking errors under actuation faults. (a)the ϕ_i 's tracking errors. (b)the θ_i 's tracking errors. (c)the ψ_i 's tracking errors.

V. Conclusions

In this article, we focus on the modeling and a generalized control framework for the coordinated quadrotor UAVs-Load transportation systems. A relatively complete dynamic models for multiple quadrotor UAVs are established under undetectable actuation faults and uncertain wind interference, the consensus control problem of how to make the UAVs maintain the designed geometric configuration is formulated correspondingly, the adaptive fault-tolerant controllers based on neighboring UAVs' state information are developed.

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